A microscope is needed to see cells.

Cells need to be small in order to work efficiently.

The cell is to biology as the atom is to chemistry – cells are the elementary structural unit of all living organisms.

New cells are formed by the division of living cells.

Cell division produces new cells for growth, repair and reproduction.

We each began life as one cell and now we are make up of 100,000,000,000,000 cells and each second of our life we make 50,000,000 new cells.

Using the microscope to view cells

- Switch on light or adjust the mirror to get good light.
- Place a small piece of onion epidermis on the glass slide.
- Place a drop of water on the plant tissue.
- Lower the cover slip gently onto the slide to cover the plant tissue.
- Place the slide on the stage of microscope.
- Turn the lowest power objective lens (X4) until it is ‘clicked’ into position.
- Turn the coarse focussing wheel to bring the lens as close to the slide as possible.
- Look through eyepiece and turn coarse focusing knob to slowly move lens away from slide until plant tissue is in focus.
- Observe the plant cells and move the slide to scan the tissue.
- Before increasing the magnification, move the area of greatest interest into the very centre of your field of view.
- Click the next higher objective lens (X10) into position.
- Only turn the fine focusing wheel to sharpen the image.
- Note the greater detail – draw a labelled diagram of part of the tissue.
- Before increasing the magnification, move the area of greatest interest into the very centre of your field of view.
- Click the highest objective lens (X40) into position.
- Only turn the fine focusing wheel to sharpen the image.
- Note the greater detail – draw a labelled diagram of a plant cell.

When finished: click the lowest power objective into position and turn the coarse focusing wheel to move the objective as far away as possible from the stage.

Iodine may be used to stain plant cells for microscopic examination.

Methylene blue is used to stain animal cells e.g. human cheek cells.

Cell structures

Structures common to plant and animal cells
Nucleus: has DNA, the genetic material; controls cell structure activity.
Chromosomes: made of DNA and protein, found in nucleus.
Gene: a particular section of DNA, which controls a definite inherited trait.
Cell membrane: surrounds the cell keeping all its contents close together; controls movement of substances in and out of cell keeping its chemistry perfect.
Cytoplasm: a complex sticky solution where many chemical reactions take place.

Structures in plant cells but not in animal cells
Cell wall: made of cellulose, strengthens and supports cell; prevents the cell bursting.
Chloroplast: photosynthesis, present in green plant cells; contains green plant pigment chlorophyll, which absorbs the light energy needed for photosynthesis.
Vacuole: stores water, food, minerals and waste; plays a support role in the soft non-woody parts of plants; the liquid of the vacuole is called cell sap.

Note: plant cells are generally much bigger than animal cells; plant cells store excess glucose as starch.

Cell types

Plants and animals contain many different types of cells, which are adapted for their function – the human body has about 300 different types of cells.

Muscle cells: contract to cause movement – move bones, move food along the gut, move blood along the blood vessels.
Nerve cells: their long fibres carry messages directly and quickly to their target.
Red blood corpuscles: contain haemoglobin to transport oxygen efficiently.
White blood cell: give protection against disease causing organisms and cancer.

Leaf cells: contain chloroplasts for photosynthesis.
Root hair cells: have large surface area to take up water and minerals from soil.
Parenchyma cells: store starch.
Epidermal cells: protect the surface of leaves.

Tissues

A tissue is a group of similar cells specialised to carry out a particular job.

Muscle cells: form muscle tissue for movement.
Nerve cells: form nerve tissue for rapid transfer of information.
Bone: plays an important role in protection, support and movement in the body.
Blood: transports materials throughout the body – can be considered to be a complex tissue as it has a number of different cell types.

Xylem tissue transports water in a plant.
Phloem tissue transports food in a plant.
Photosynthetic tissue in a leaf produces food for the plant.
Parenchyma is storage tissue and holds food in reserve for later use.

Organs

An organ is a group of tissues working together to carry out a particular job.

Heart: a mixture of muscle and nervous tissue – its job is to pump the blood.
Lungs: their job is to absorb oxygen and excrete carbon dioxide.
Stomach: its job is to digest protein plus absorb water, glucose and minerals.
Brain: its job is to control the body's activities, think, learn and make decisions.

Root: anchorage, food storage, water and mineral absorption are its main jobs.
Stem: support the leaves, buds and flowers; food storage,
Leaf: photosynthesis and transpiration.
Flower: sexual reproduction – pollination, seed formation and seed dispersal.

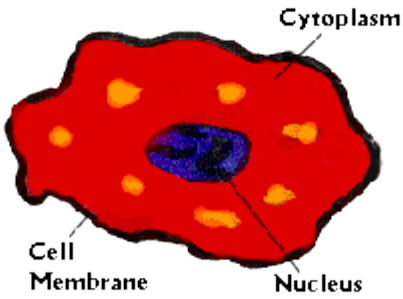
Organs cannot work by themselves; they depend on other organs to supply things they need.

Systems

A system is a group of organs working together to carry out a particular job.

Digestive: oesophagus, stomach, liver, pancreas, small intestine, large intestine.
Nervous: brain, spinal cord, nerves, eyes, ears, taste buds, nose, skin, nerves.
Circulatory: heart, arteries, veins, blood.

Animal cells are the cells found in animals. You are made up of trillions of animal cells.
These are the basic parts of an animal cell:



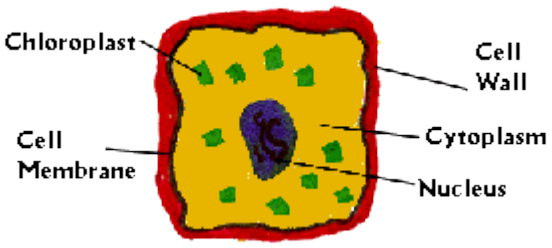
Cell Membrane: A cell membrane works like a screen. It lets the good things in like nutrients and lets the bad things out like waste. It is like a sieve.

Cytoplasm: Cytoplasm is the jelly-like part of the cell where the action takes place. It is where the nutrients are used.

Nucleus: The nucleus is the brain of the cell. It controls the cell, and tells it what to do. The nucleus also contains the DNA which is like a blueprint. A blueprint is a plan that people use when they build. A cell's blueprint is the plan the cell uses when it reproduces.

One of the most important ways that animal cells are different from plant cells is that animal cells can't make their own food. Plant cells make their own food in a process called photosynthesis. Animal cells can't do that. Instead, animals must eat food. The animal cells use the food to make the energy they need to reproduce and carry out basic functions. This process takes place in a part of the cell called the mitochondrion.

If you would like to see an actual photograph of a pig cell or of mitochondrion viewed through the lens of an electron microscope, visit the Nanoworld Image Gallery (you'll need to register in order to view the gallery).



Plant cells are cells that are in plants. Plant cells are like animal cells, but they have a cell wall and chloroplasts. Here's a labeled picture of a

plant cell. You will see all the main parts of the cell. Listed below are definitions of those words. Read on to learn more about photosynthesis.

Cell Membrane: A thin layer that surrounds the cell.
Cell Wall: A rigid layer that surrounds the cell membrane of a plant cell.
Chloroplast: (KLOR-o-plast) A cell structure in which photosynthesis takes place, and it contains the chloropyll.
Chlorophyll: (KLOR-a-fil) A green substance in plant cells that helps to make food.
Cytoplasm: (SIGH-ta-plaz-em) The jellylike liquid in cells where activities take place.
Nucleus: (NEW-kle-us) The dark structure inside the cell that controls the cell's activities and contains material such as DNA.
Photosynthesis: (foto-SIN-thi-sis) The process when plants use lights energy to make food.

A very special process that takes place inside of the **chloroplasts** in plant cells is called **photosynthesis** (foto-SIN-thi-sis). Photo means "light" and synthesis means " putting things together." In photosynthesis, green plants actually make their own food. That is so cool! Imagine if we could make our own food. I wonder what we'd make? Personally, I would make pizza. Plants seem to have the right idea since they make sugar. This amazing process is responsible for everything we eat. That's because animals eat plants, and we eat the animals and the plants.

If you would like to see an actual photograph of a chloroplast viewed through the lens of an electron microscope, visit the Nanoworld Image Gallery (you'll need to register in order to view the gallery).

In photosynthesis, the **chloropyll** (KLOR-a-fil) takes in energy from sunlight. The energy comes in the form of a tiny bundle known as a photon. The photon hits a molecule of water inside the chlorophyll. The photon's energy splits the water molecule into hydrogen and oxygen. The hydrogen combines with carbon dioxide (which the plant has absorbed from the air) to make sugars or glucose. The oxygen is released back into the atmosphere to give us more air. Did you know that the largest suppliers of oxygen in the whole world are tiny plants known as photoplankton? They are found in the oceans and are really important to giving us the oxygen needed to keep life going on our planet.

Believe it or not, a plant uses only about one-sixth of the energy it gets from the sun to nourish itself. The rest of the energy is stored in the glucose until it is eaten by other animals or humans. What an amazing